1. Reproduction

1.1. Asexual Reproduction

Asexual Reproduction: the process resulting in the production of genetically identical offspring from one parent.

- Bacteria:
  - Reproduced by binary fission, each bacterium divides into two.
  - The generation time is the time taken for a cell to divide into 2.

Advantages	Disadvantages
Fast: no need to find a mate, fertilise, etc.	No genetic variation/biodiversity
Good characteristics are kept	Harmful genes transferred
Do not need to carry offspring	Overcrowding: fighting for food/competition
No pollinators	Prone to extinction, disease, environmental change

1.2. Sexual Reproduction

Sexual reproduction: a process involving the fusion of the nuclei of two gametes (sex cells) to form a zygote and the production of offspring that are genetically different from each other

- Fertilisation: the fusion of gamete nuclei
- The nuclei of gametes are haploid, and the nucleus of a zygote is diploid
- Diploid: Full Set of Chromosomes
- Haploid: Half Set of Chromosomes

Advantages	Disadvantages
Produces genetically different offspring	It takes lots of time and energy
Reduced risk of extinction	Mate required
	Energy on improving appearances or pollen volume for pollination (plants)

1.3. Parts of a Flowering Plant

- Flowers are the reproductive organ of the plant
- They usually contain both male and female reproductive parts
- The structure of insect and wind-pollinated flowers are slightly different as each is adapted for their specific function

Insect Pollinated Flowers

Wind Pollinated Flowers

Functions

- Sepal: protect the flower bud.
- Petal: brightly coloured and scented and may have nectars, which are all used to attract insects. Petals in wind-pollinated flowers are tiny and used for pushing the bracts (leaf-like structures) apart from exposing stamens and stigma.
- Anther: has pollen sacs with pollen grains that contain the male nucleus (male gamete).
- Stigma: platform on which pollen grains land
- Ovary: hollow chamber, ovules grow from the walls.

1.4. Pollination

Pollination: transfer of pollen grains from the male part of the plant (anther of stamen) to the female part of the plant (stigma).

- Agents of pollination: insects, birds, mammals, water and wind
- Fertilisation occurs when a pollen nucleus fuses with a nucleus in an ovule

Insect Pollinated	Wind Pollinated
Bright, colourful petals – attract insects	Dull petals
Sweetly scented	No scent
Contains nectar	No nectaries
A moderate amount of pollen	Huge amount of pollen
Pollen is spiky/sticky, large in size	Pollen round and smooth, Small and light

- Pollen tube: pollen grain lands on the stigma and creates a tunnel down the style, through the micropyle to ovules.
- Ovule: seed
- Ovary: fruit

Self Pollination

Self Pollination: the transfer of pollen grains from the anther of a flower to the stigma of the same flower or a different flower on the same plant.

Advantages	Disadvantages
Genetically identical	Lack of genetic variation
High chance of successful pollination	Increases competition between plants
Fast and saves time	Susceptible to the same disease

Cross-Pollination

Cross-pollination: the transfer of pollen grains from the anther of a flower to the stigma of a flower on a different plant of the same species.

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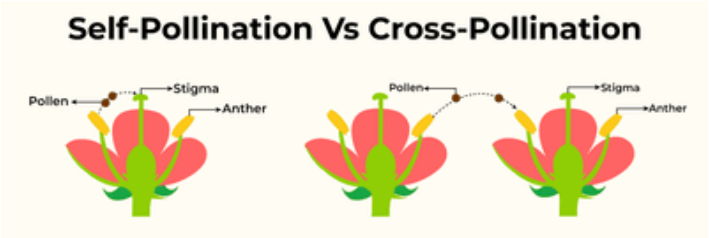
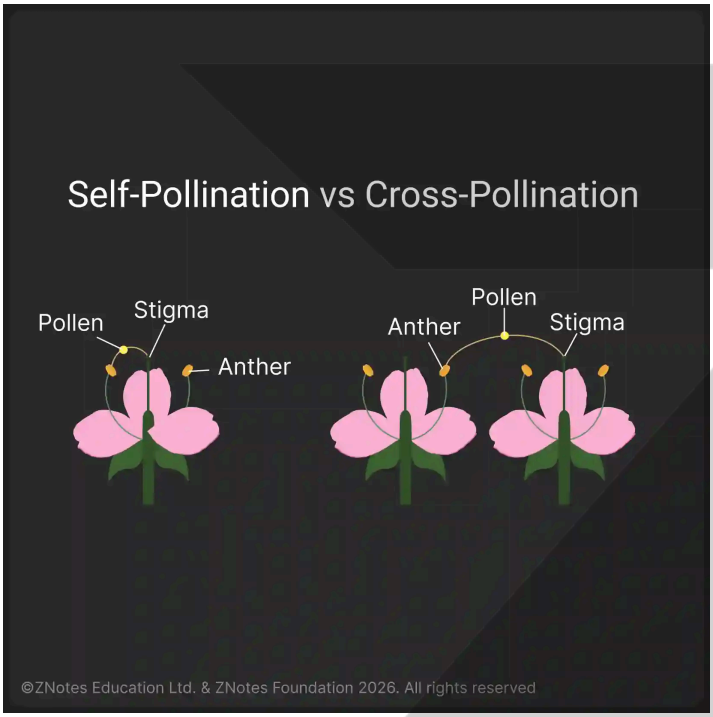
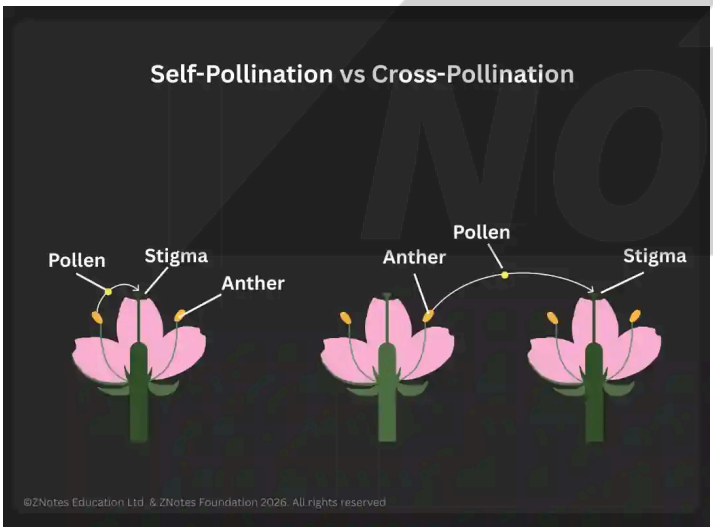
Advantages	Disadvantages
Increases variation	Reliance on pollinators
Quick to adapt to surroundings	Wastage of pollen
Less susceptible to diseases	More energy required







1.5. Fertilisation in Plants

- Pollen tube growing from a pollen grain
- Pollen (tube/grain) releases enzymes
- The pollen tube then grows down the style/grows into the ovary
- Pollen nucleus travels down the pollen tube
- Pollen tube grows and moves through the micropyle into the ovule
- Fertilisation occurs
- The pollen nucleus fuses with the female nucleus
- A zygote is formed

1.6. Germination

Germination: A process controlled by enzymes

- **Water:** activates enzymes to turn insoluble food stores into soluble substances, and makes tissues swell so that the testa splits
- **Oxygen:** enters through the gaps in the testa (along with water), and is used in aerobic respiration.
- **Temperature:** must be suitable for enzymes to work (at optimum temperature).

1.7. Sexual Reproduction In Humans

Male Reproductive System

- **Testes:** have many coiled tubes that produce sperm, and the cells between tubes produce testosterone.
- **Scrotum:** holds testicles
- **Sperm duct:** carries sperm from testicles to urethra.
- **Prostate gland:** makes seminal fluid
- **Urethra:** carries semen from the sperm duct to the tip of the penis
- **Penis:** male sex organ used to transfer semen to the female.

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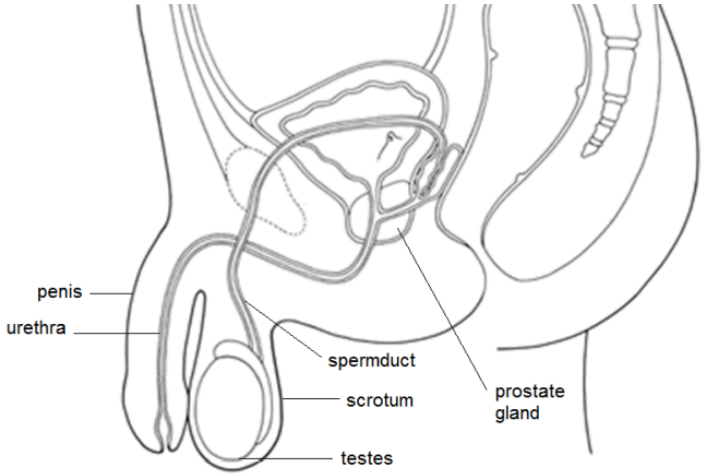
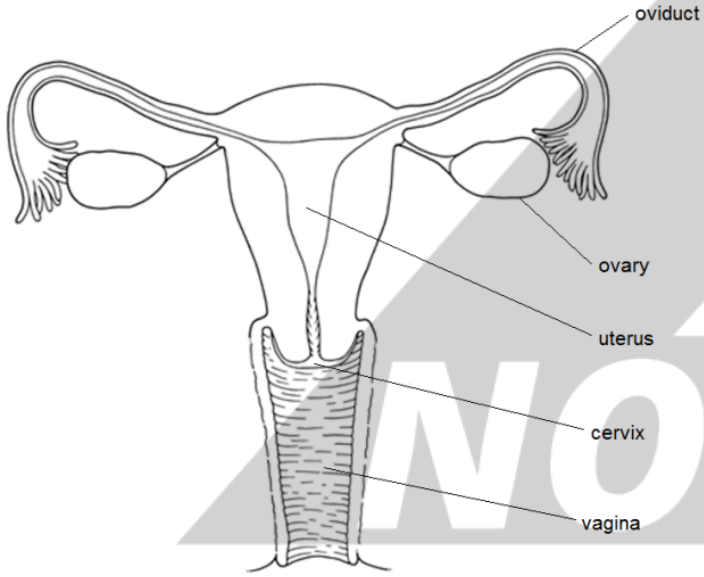


Female Reproductive System

- **Ovary:** contains follicles that develop into the ova and produces progesterone and oestrogen
- **Oviduct (fallopian tube):** carries the ovum to the uterus
- **Uterus (womb):** where the fetus develops.
- **Cervix:** neck of the uterus: a robust and rigid muscle, moist by mucus with a small opening
- **Vagina:** receives the penis during intercourse and way out for baby at birth. Moist tube of muscle, flexible and secretes mucus



1.10. Fertilisation and Early Development

Fertilisation: The fusion of the nuclei from a male gamete (sperm) and a female gamete (egg cell).

- **Development of a Zygote:**
  - One sperm penetrates
  - The ovum membrane alters to form a barrier against sperm
  - The head of the sperm (male nucleus) approaches and then fuses with the nucleus of the ovum.
  - The zygote divides over and over to make a ball of cells called an embryo.
  - It implants itself into the wall of the uterus (implantation).
- **Development of fetus:** The zygote is changed through growth (mitosis) and development (organisation of cells into tissues and organs forming an embryo and eventually a fetus.)
- **Umbilical cord:** contains the umbilical artery, which carries deoxygenated blood and waste products from the fetus to the placenta and the umbilical vein, which carries oxygenated blood and soluble food from the placenta to the fetus. (Contains fetus' blood)
- **Placenta:**
  - An organ that forms during pregnancy
  - Allows the exchange of soluble materials such as oxygen, nutrients, and waste products between the mother's blood and the fetus's blood.
  - Provides physical attachment between uterus and fetus. (Contains mother's blood)
- **Amniotic Sac:** membrane which encloses amniotic fluid, broken at birth.
- **Amniotic Fluid:** protects the fetus against mechanical shock, drying out and temperature fluctuations
- Some pathogens and toxins can pass across the placenta and affect the fetus.

1.11. Adaptive Features of Gametes

Sperm (Male Gamete)

1. Small in size
2. Elongated and streamlined with energy storage

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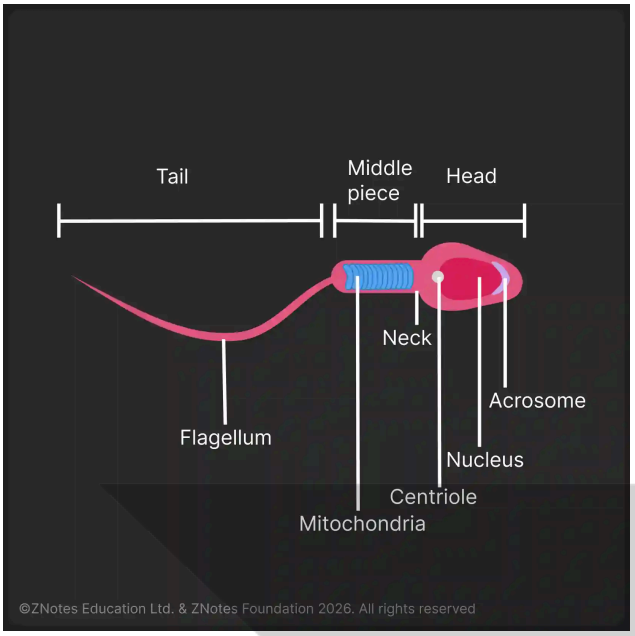
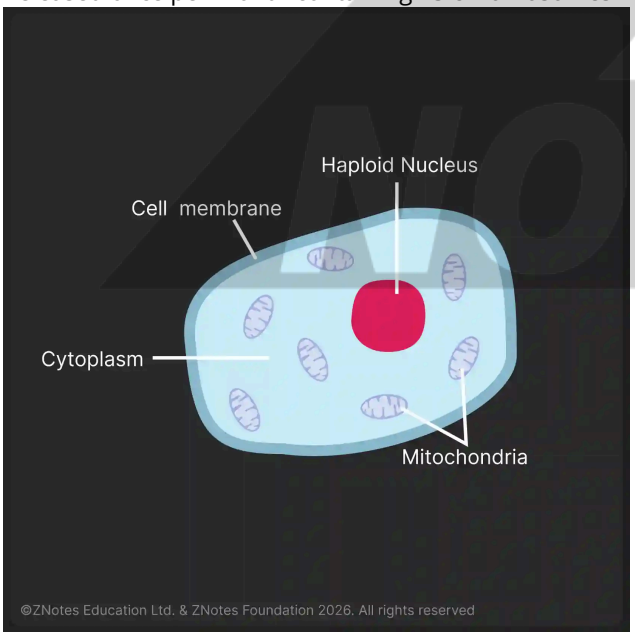
3. Millions in numbers containing 23 chromosomes



Features	Functions
Flagellum	Propels the sperm to swim
Mitochondria	Respiration to release energy for swimming
Enzymes in the acrosome	Release digestive enzymes to digest the jelly coat

Egg Cell (Female Gamete)

1. Larger in size
2. Spherical protein/fat in the cytoplasm
3. Moves with the help of Cillia
4. Released once per month containing 23 chromosomes



Features	Functions
Energy storage	Development of zygote
Jelly coat	Changes at fertilisation

1.12. Sexual Hormones in Humans

The roles of testosterone and oestrogen in the development and regulation of secondary sexual characteristics during puberty

- **Primary sexual characteristics**: present during development in the uterus and are the differences in reproductive organs etc., between males and females
- **Secondary sexual characteristics**: are the **changes that occur during puberty** as children become adolescents
- At puberty, the pituitary gland starts to stimulate the primary sex organs: the testes in males and the ovaries in females.
- They only affect the target organs, which have receptors that can recognize them.
- Causes secondary sexual characteristics such as the growth of pubic hair and maturation of sexual organs.

1.13. The Menstrual Cycle

Day 1 to 5:

- In the ovary, *FSH* secreted by the *Pituitary Gland* to stimulate the maturation of ONE *follicle* in the ovary.
- In the uterus: the *endometrium* breaks down; menstruation
- Day 5 to 12:
  - In the ovary, the follicle keeps maturing
  - In the uterus, *oestrogen* is secreted by *follicle* and the ovarian tissues to prepare the *endometrium*
- Day 13/14/15:
  - In the ovary, *LH* is also secreted by the *Pituitary Gland* to trigger the release of the egg from the follicle into the fallopian tube. Ovulation happens on **Day 14**.
- Day 15 to 28:
  - In the ovary, LH causes the empty follicle to turn into a Corpus Luteum.
  - The Corpus Luteum produces progesterone, a hormone
  - In the uterus, progesterone keeps the lining (endometrium) thick, in case a fertilised egg (embryo) needs to implant.
- Day 28 – Scenario 1: Egg not fertilised
  - No implantation takes place, and the *Corpus Luteum* degenerates, causing a lack of progesterone.
  - This means that *endometrium* is no longer kept thick, the cycle returns to Day 1
- Day 28 – Scenario 2: The egg is fertilised
  - Implantation occurs.
  - This makes the hormones keep the *Corpus Luteum* maintained, which means that *progesterone* is high.
  - This keeps the *Endometrium* thick for pregnancy

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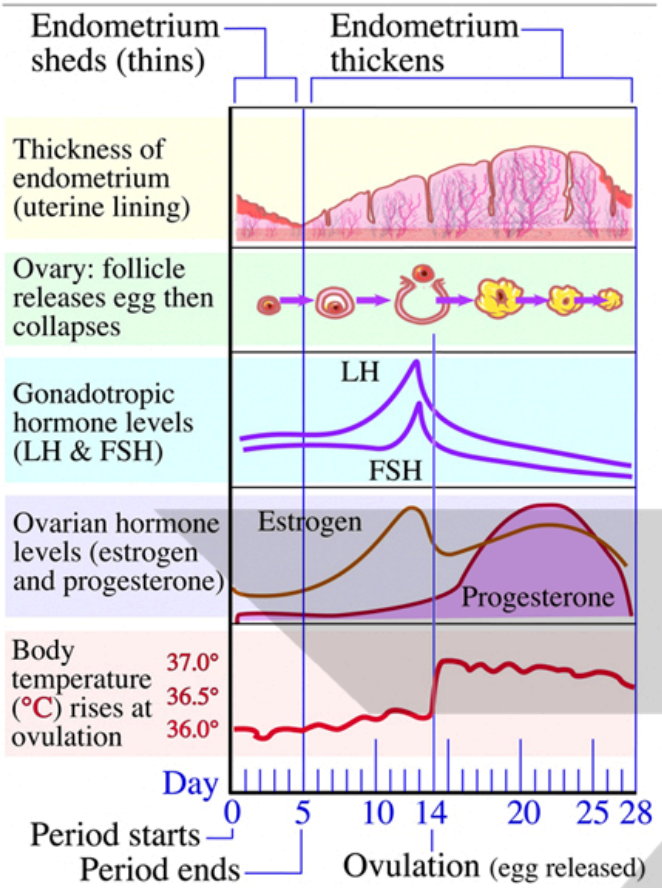

1.15. Hormones in Menstrual Cycle

- **Oestrogen**: secreted by the ovaries. It stops FSH from being produced - so that only one egg matures in a cycle, and it stimulates the pituitary gland to release the hormone LH.
- **Progesterone:** secreted by the **placenta** during pregnancy, while during the menstrual cycle is by the **ovaries**. Its primary function is maintaining the uterus lining during the middle of the menstrual cycle and pregnancy.
- **Follicle-stimulating hormone (FSH):** secreted by the pituitary gland. It causes an egg to mature in an ovary and stimulates ovaries to release oestrogen hormone.
- **Luteinizing hormone (LH):** secreted by the pituitary gland and causes mature eggs to be released from the ovary.

1.16. Sexually Transmitted Infections

Human Immunodeficiency virus (**HIV**) is one example of a sexually transmitted infection.

- **Transmission:** Intercourse, blood transfusion, organ transplant or sharing a needle with an infected person
- **How it affects the immune system:**
  - Infects and destroys lymphocytes
  - Decreases the efficiency of the immune system
  - The body becomes liable to infection by other pathogens
  - This may lead to AIDS and death from infection

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